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ADVANCED DATA MANAGEMENT 2025/2026

July 16, 2026

CLOSED BOOK

Q1. (up to 10 lines) [**only for COMPUTER ENGINEERING STUDENTS**]

Explain the *data locality* principle of large-scale data systems and provide at least one example.

Q2. (up to 10 lines)

With reference to the three-step design methodology, explain why representing selection attributes as nested attributes in the schema could cause problems in Cassandra. Use an example if useful.

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Q3. (up to 10 lines)

Discuss the notion of index-free adjacency in graph databases and its implications.

Q4. (up to 10 lines) **[only for COMPUTER SCIENCE STUDENTS]**

Discuss the main motivations and use cases for vector databases.

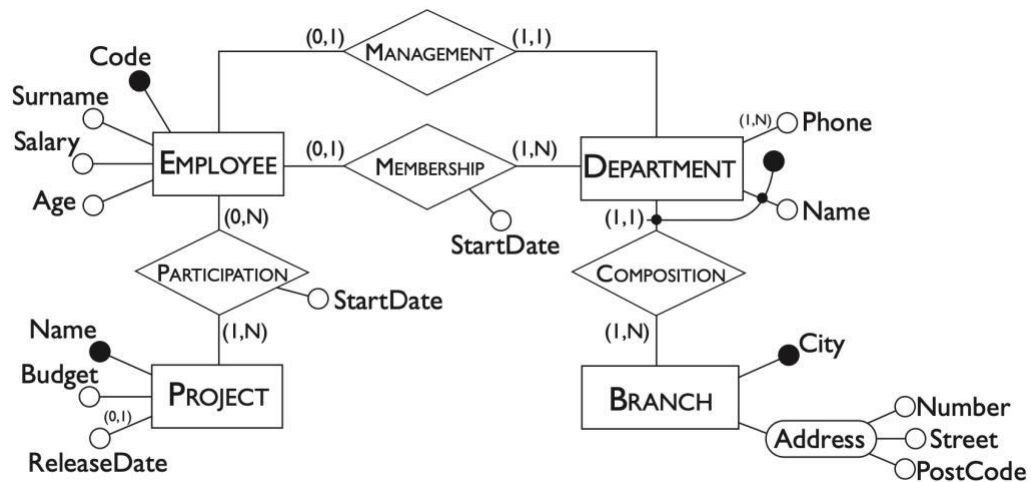
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Exercise 1 [NoSQL aggregate-oriented data stores]

Consider the following entity relationship diagram, representing the departments of a company with many branches.



The workload is the following:

- **Query 1:** Retrieve the city and the address of the branches having at least one employee younger than 24 years old.
- **Query 2:** Retrieve all information about employees who are managers of a department located in Genova.
- **Query 3:** Retrieve all the branches including a department with a certain name.

Based on the previous ER schema and workload, answer the following questions, **motivating all your choices**.

1. Use the aggregate-oriented design methodology (STEP 1-2-3) to design a set of logical aggregates, in meta-notation, for the given domain and the selected workload. Present the output of each step of the methodology.

Queries

Q1

Q2

Q3

Annotation: please annotate the input ER diagram

Aggregates

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2. Revise and extend the logical aggregate schemas to consider partition keys and super-identifier information.
3. For each aggregate-oriented schema and each associated query, specify the query type (either batch or random-access) and whether it can be efficiently executed on the collection (for random-access queries only), without creating any index.
4. Determine which indexes (if any) should be created to improve the efficiency of workload queries.
5. Suppose to implement the designed aggregates in Cassandra.
 - a. Explain if the designed schemas should be modified to improve query execution. If the answer is yes, present the new schemas in meta-notation.
 - b. For each aggregate schema, present the PRIMARY KEY clause of the corresponding table in Cassandra and explain which indexes should be created (if any).
 - c. Specify Query 1 in CQL and point out how many nodes will be accessed during the execution.

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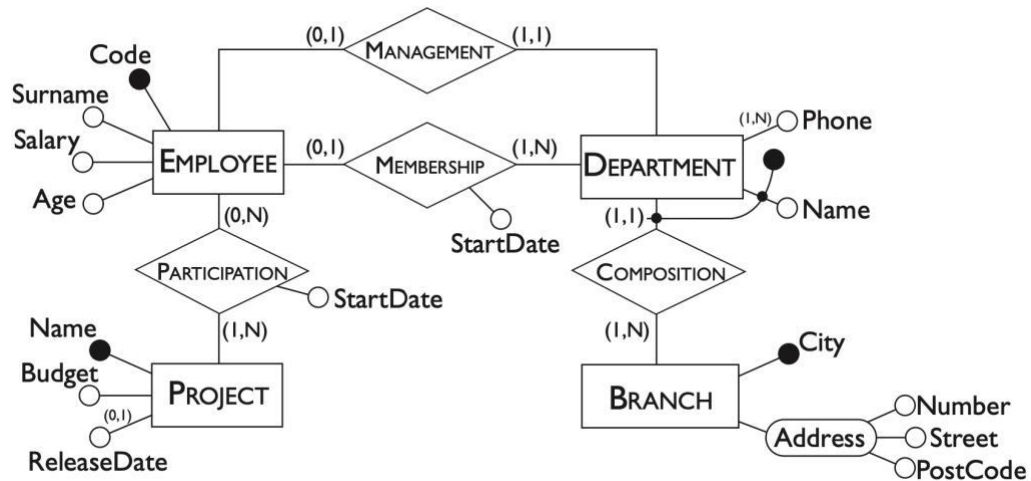
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Exercise 2 [NoSQL graph-based data stores]

Consider the following entity relationship diagram, and assume now that:

- employee can be members of multiple departments, with an additional attribute **allocation_percentage**
- projects may involve multiple departments
- **participation** relationship should take into account and represent the participation of employees in projects on behalf of a specific department.



1. Present the most suitable logical schema in a graph-based data store relying on labelled property graphs, like Neo4j (in the form of a graph pattern or of a sample instance), for the domain described by the scenario above (ER diagram modified as discussed in the text) including some instances for each node label/edge type.

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2. Assuming the information above is represented using a labelled property graph (Neo4j), write a Cypher query that, for a given project with name “DAMA” returns the departments involved in the project, and for each of them the number of distinct employees working on the project on behalf of the department, and the average percentage of allocation percentage of the those employees to that department.

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Exercise 3 [Data management for AI driven application] [only for COMPUTER SCIENCE STUDENTS]

With reference to scenario discussed in Exercise 2.1 for LPG:

1. Consider one small example instance of your choice (containing at least an instance of the relationships) and discuss the differences in how the relationships are represented in a labelled property graph (e.g., Neo4j) and in RDF. Your answer should show the graphs, clearly explain and compare the modeling choices adopted in each case.

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2. Assume now that the information of the employee-project-department scenario above is represented as a graph (e.g., a labelled property graph or an RDF graph). Discuss whether and how such a representation could be exploited in a Retrieval-Augmented Generation (RAG) setting involving Large Language Models (LLMs). In particular,
 - (a) Describe what unstructured data can be relevant in that context. Specifically, in addition to the graph-structured data above specify what objects are embedded, chunking strategy, metadata schema. You may use diagrams, tables, or pseudo-schema definitions.
 - (b) With reference to this scenario, show an example of request that exploits both the Graph and the RAG component of the architecture. Highlight the parts that can be executed exactly and those that will be approximated.